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DEER STAND CLIMBING SYSTEM

Abstract

A hunting platform system including a platform couplable to a tree and a band extending around the tree. The band being connected to both sides of the platform. The platform has a heel hole positioned in the platform to accommodate a heel of a hunter.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This is a non-provisional patent application based upon U.S. provisional patent application Ser. No. 63/558,932, entitled “DEER STAND CLIMBING SYSTEM”, filed Feb. 28, 2024.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to a tree climbing system by way of the invention that additionally serves as a platform coupled to the tree for a hunter to supported.

2. Description of the Related Art

[0003] A tree stand is a platform used by hunters to elevate themselves above the ground, typically attached to a tree. This elevated position provides a better vantage point for spotting deer and other game, and it helps to keep the hunter's scent above the ground, reducing the chance of detection by the animal.

[0004] There are several types of hunting stands such as ladder stands, climbing stands, and hang-on stands. Ladder stands have a ladder attached to the platform, making them easy to climb. They are stable making them suitable for long hunting sessions. Ladder stands are typically coupled to a tree using straps. Climbing stands are portable and allow hunters to climb the tree using the stand itself. They are ideal for hunters who need to move to locations frequently, or that can't leave a stand attached for days or weeks. Hang-on stands are lightweight and can be hung on trees using straps or chains. They require separate climbing aids like climbing sticks or steps.

[0005] The benefits of using a tree stand include improved visibility, reduced scent detection and safety. Improved visibility results from being elevated, which gives hunters a better view of the surrounding area, making it easier to spot animals as they move on the ground. Reduced scent detection is inherent to being in an elevated position, which helps keep the hunter's scent above the ground, making it less likely for the quarry to detect them. Further, tree stands can provide a safer shooting angle, reducing the risk of stray bullets causing damage.

[0006] What is needed in the art, is a climbing hunting platform with features that increase the climbing efficiency, promote comfort and ensure enhanced footing.

SUMMARY OF THE INVENTION

[0007] The present invention provides an improved hunting platform.

[0008] The invention in one form is directed to a hunting platform system including a platform couplable to a tree and a band extending around the tree. The band being connected to both sides of the platform. The platform having a heel hole positioned in the platform to accommodate a heel of a hunter.

[0009] The invention in another form is directed to a platform couplable to a tree including a band extending around the tree and connected to both sides of the platform; and at least one foot loop coupled to the platform. The at least one foot loop being used by the hunter to lift or lower the system on the tree.

[0010] The invention in yet another form is directed to a method of a hunter using a hunting platform system, the method including the steps of positioning, connecting, climbing, inserting, lifting, engaging and repeating. The positioning step including the positioning of a hunting platform against a tree. The connecting step including the connecting of a band to two sides of the platform and extending around the tree. The climbing step including the hunter climbing upon the hunting platform. The inserting step including the inserting of the hunter's feet into at least one foot loop. The lifting step including the lifting of the hunting platform with the hunter's feet while the hunter is otherwise supported. The engaging step including the engaging of the tree with the platform at a new position. The repeating step including the repeating of the lifting and engaging steps until the platform is at a desired height.

[0011] Advantageously the present invention provides a climbing stand with a heel hole feature, thereby enhancing the performance of the hunting stand.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The above-mentioned and other features and advantages of this invention, and the [0013] manner of attaining them, will become more apparent and the invention will be better understood by reference to the following description of embodiments of the invention taken in conjunction with the accompanying drawings, wherein:

[0014] FIG. **1** illustrates, a hunting platform system of the present invention engaged with the trunk of a tree;

[0015] FIG. **2** is a top down perspective view of the hunting platform system of FIG. **1**;

[0016] FIG. **3** illustrates the hunting platform system of FIGS. **1** and **2** in a perspective view from the tree side;

[0017] FIG. **4** illustrates an end view of a memory band of the present invention used as a part of the hunting platform system of FIGS. **1-3**;

[0018] FIG. **5** illustrates a latching system of the present invention that interfaces with the memory band of FIG. **4**;

[0019] FIG. **6** is a cross-sectional view of the latching system of FIG. **5** and is part of the hunting platform system of the previous Figs.;

[0020] FIG. **7** is another view of the hunting platform system of the previous Figs., particularly illustrating the use of a heel hole in the floor of the platform; and

[0021] FIG. **8** is a view from beneath the platform showing the deep engagement of the hunter's heel in the heel hole previously illustrated.

[0022] Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate embodiments of the invention and such exemplifications are not to be construed as limiting the scope of the invention in any manner.

DETAILED DESCRIPTION OF THE INVENTION

[0023] Referring now to the drawings, and more particularly to FIGS. **1** and **2** there is shown a deer stand climbing system **10** having a body **12**, a memory band **14**, a pivoting arm **16**, a pivoting arm strap **18** and a web ladder **20**. Body **12** has a platform **22**, angled side members **24**, foot loops **26**, and band latches **28**.

[0024] Platform **22** as illustrated has an open pattern of structural features that provide rigidity yet light weight. Platform **22** advantageously has a heel hole **H** that is shaped to accommodate the heel of a hunter to thereby provide a certain and robust control of the hunter's position while using a saddle/sling. Angled side members **24** extend from a front edge of platform **22** at an angle that advantageously allows for the interaction of the back edge of platform **22** that engages the side of the tree and memory band **14** that extends around the back side of the tree so that system **10** will support the weight of the hunter standing on platform **22**. Side members **24** have connections on the sides of platform **22** that provide a quasi-triangular rigidity of members **24** to platform **22**.

When the operator of system **10** had reached a desired height on the tree, the operator pivots pivoting arm **16** upward against the tree, as shown in FIG. **1**, and couples strap **18** to secure arm **16** to the tree, thereby preventing system **10** from disengaging from the tree. Pivoting arm **16** serves as a pawl **16** or a dog **16** keeping platform **22** coupled to the tree.

[0025] Memory band **14** is a resilient band that has positional memory, so that when band **14** is disengaged on one side by a release of a band latch **28** and band **14** is moved by the operator to go around the tree, the disengaged end of memory band **14** returns to a position proximate to the band latch **28** so that memory band **14** can be easily reengaged with that band latch **28**.

[0026] The operator can initially position system **10**, as illustrated in FIG. **1** a few feet from the ground, with the exception that strap **18** would not be engaged around the tree and pivoting arm **16** would be pivoted downward toward platform **22**. The operator can then climb web ladder **20** and stand on platform **22**. At this point the operator will affix an end of a sling or swing cord or rope (not shown, but known to hunters) around the tree at an elevated location on the tree. The sling has

a portion upon which the operator can sit and which serves to support the operator's weight. The operator then inserts his/her feet into foot loops **26** and lifts system **10** upward along the trunk of the tree, while the operator's weight is supported by the sling. The tops of the operator's feet lift against foot loops **26** to lift system **10** upward. Advantageously, foot loops **26** are ridged to ensure that they can be easily engaged and disengaged by the operator. This ensures that the operator is not unduly coupled to platform **22** and can easily separate therefrom. The operator then steps down on platform **22** causing system **10** to engage with the tree and support his/her weight. The operator then repositions the sling cord to a higher position on the tree and repeats that action to raise system **10** to a higher position on the tree.

[0027] Once the operator has reached a desired height then pivoting arm **16** is engaged with the tree and strap **18** is used to secure arm **16** against the tree. The operator then positions the sling cord to provide a preferred comfort and uses platform **22** to position him/herself about the tree as is needed for the intended viewing from the elevated position established on the tree.

[0028] When the operator is finished the operation can be reversed to return to ground level, or the operator can disconnect system **10** and rappel down the side of the tree using his/her feet to “walk” down the side of the tree. The sling mechanism, or a rappelling system, which is not part of the present invention, includes devices that disconnect the rope/cord from the tree so that nothing is left connected to the tree and the operator can simply carry the sling and system **10** from the wooded area. System **10** is advantageously coupled to a backpack arrangement for easy transport into and from the wooded area.

[0029] Now, additionally referring to FIG. **3** there is shown another view of band **14** having a series of spaced apart holes **30**, which allow band latches **28** to adjust the effective length of band **12** about the tree. It is also contemplated to provide differing lengths of band **12** for differing diameters of trees. It is further contemplated to provide altered spacing of holes **30**. The advantage of the resilient nature of memory band **14** is that it holds its form to allow for easy latching and to clear the tree as system **10** is being raised or lowered along the trunk of the tree.

[0030] Now, additionally referring to FIG. **4**, band **14** is shown in cross-section having a thickness T and a width W. Thickness T may be $\frac{1}{4}$ of an inch and width W may be $\frac{3}{4}$ inch. The approximate three to one ratio of width W to thickness T has been discovered to help provide optimal memory, compactness and strength for band **14**.

[0031] Now, additionally referring to FIG. **5**, latch **28** is illustrated showing that a pulling motion releases a pin (shown in FIG. **6**) from a hole **30** allowing band **14** to be adjusted or removed from latch **28**. Latch **28** is integral with angled sides **24** thereby establishing the desired angle of memory band **14** relative to platform **22**.

[0032] Now, additionally referring to FIG. **6**, there is illustrated some of the interior features of latch **28** with pin P shown going through one of the holes **30** in band **14**. Other latching mechanisms are contemplated such as levers. It is also contemplated that band **14** can have surface features that would allow a latching mechanism to interface with the surface features to latch band **14** to the latch mechanism without the need for holes **30**.

[0033] Now, additionally referring to FIGS. **7** and **8**, the use of heel hole H can be seen from an upper perspective and a lower perspective. As can be seen the heel of a boot or shoe fits from any angle into heel hole H. Heel hole H is quasi-circular or has a D-shape with the flat portion being horizontally arranged perpendicular relative to a longitudinal extent of the tree.

[0034] Advantageously, as the hunter/operator is sitting in a sling or swing, he/she can be positioned against the tree and securely establish the position by having their heel in heel hole H. This provides for better and more stable positioning while their hands are free to use their weapon or camera. This results in more accuracy and a better hunting experience. This avoids losing a desired positioning as the tread of the platform and their shoe is inadequate to prevent them from sliding, particularly, since the swing/sling allows positioning of the legs of the hunter at angles that are significantly removed from being vertical. Heel hole H being sized and shaped so that at least a

part of the heel of the hunter's footwear can extend through a bottom plane of platform 22.

[0035] Although, the terms 'operator' and 'hunter' are used herein, they are meant to refer to whoever is operating and using system 10, and can include anyone who is using system 10, whether system 10 is ultimately used for hunting, for observing or any other purpose.

[0036] While this invention has been described with respect to at least one embodiment, the present invention can be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains.

Claims

1. A hunting platform system, comprising: a platform couplable to a tree; and a band extending around the tree and connected to both sides of the platform; wherein the platform has a heel hole positioned in the platform to accommodate a heel of a hunter's footwear.
2. The hunting platform system of claim 1, further comprising at least one foot loop coupled to the platform, the at least one foot loop being used by the hunter to lift or lower the system on the tree.
3. The hunting platform system of claim 2, wherein the at least one foot loop is rigid.
4. The hunting platform system of claim 3, wherein the at least one foot loop is two foot loops.
5. The hunting platform system of claim 1, wherein the band is a memory band having a resilient memory.
6. The hunting platform system of claim 5, further comprising at least one band latch, the memory band having a series of holes therethrough, the band latch having a pin extending through a selected hole of the series of holes.
7. The hunting platform system of claim 5, wherein the band has a thickness and a width, the width being approximately 3 times larger than the thickness.
8. The hunting platform system of claim 1, further comprising a web ladder coupled to an edge of the platform.
9. A hunting platform system, comprising: a platform couplable to a tree; a band extending around the tree and connected to both sides of the platform; and at least one foot loop coupled to the platform, the at least one foot loop being used by the hunter to lift or lower the system on the tree.
10. The hunting platform system of claim 9, wherein the at least one foot loop is rigid.
11. The hunting platform system of claim 10, wherein the at least one foot loop is two foot loops.
12. The hunting platform system of claim 9, wherein the platform has a heel hole positioned in the platform to accommodate a heel of a hunter being inserted therein, the heel hole being D-shaped.
13. The hunting platform system of claim 9, wherein the band is a memory band having a resilient memory.
14. The hunting platform system of claim 13, further comprising at least one band latch, the memory band having a series of holes therethrough, the band latch having a pin extending through a selected hole of the series of holes in the memory band.
15. The hunting platform system of claim 13, wherein the memory band has a thickness and a width, the width being approximately 3 times larger than the thickness.
16. The hunting platform system of claim 9, further comprising a web ladder coupled to an edge of the platform.
17. A method of positioning a hunting platform system, the method comprising the steps of a hunter: positioning a hunting platform against a tree; connecting a band to two sides of the platform, the band extending around a side of the tree; climbing upon the hunting platform; inserting the hunter's feet into at least one foot loop; lifting the hunting platform with the hunter's feet while the hunter is otherwise supported; engaging the tree with the platform at a new position; and repeating the lifting and engaging steps until the platform is at a desired height.

- 18.** The method of using the hunting platform system of claim 17, further comprising the step of while otherwise supported inserting the hunter's heel into a heel hole on the platform and swinging to a desired hunting position.
- 19.** The hunting platform system of claim 17, wherein the band is a memory band having a series of holes therethrough.
- 20.** The hunting platform system of claim 17, wherein the at least one foot loop is two foot loops that are rigid.
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